

Final Assignment in Data Analysis for Sustainable Development: Theory and Applications

What shapes Animal-Sourced Food Consumption?

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1 Introduction

Animal-sourced foods (ASFs) are a highly debated component of human diets. Their consumption, particularly of red and processed meats, is associated with increased health risks of cancer and cardiovascular disease[13][17]. Yet they also provide essential nutrients such as vitamin B12, proteins, and iron which are absent or scarce in most plant-sourced foods (PSFs) and help provide a healthy diet, they could be supplemented or PSFs could be fortified with these nutrients [15].

Further, food systems constitute a major driver of climate change. In 2015, between 25% and 34% of anthropogenic greenhouse gas (GHG) emissions originated from food related systems. These different estimations exist, as Crippa et al. include consumer-side emissions such as cooking and attribute all changes of land-use to agriculture, whereas Poore and Nemecek only attribute 60% of land-use[20][6]. The true share of GHG emissions probably is somewhere in between, but undeniably it plays a mayor role in climate change.

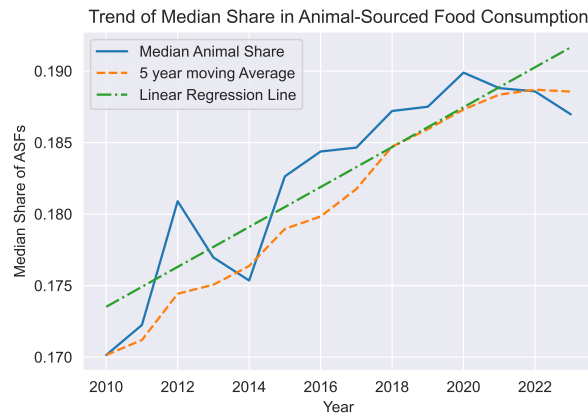


Figure 1: Median animal share per year with trend and moving average

Clark et al. argue that GHG emissions from food-systems need to be cut in half to have a grater than 50% chance of limiting global warming to 1.5°C[4]. In this context, White and Hall propose to support food security, agricultural greenhouse gas emissions, or affordability objectives individually or jointly the ASFs share should approximately be between 12% and 18%[24]. However, as displayed in Figure 1, 2023 median ASF calorie intake from animal sources globally was 18.9%, with an upwards trend, which is above their proposed limits. This global trend can also be seen on the maps of Figure 3 for many individual countries.

The climate implications of this pattern are substantial. After excluding non-food biomass use and production, ASFs account for 66.3% of food-related GHG emissions, which is in stark contrast to its share in caloric contribution. This implies 100 kcal of ASF generate 8.43 times the GHG emissions of 100 kcal PSF[27]. These numbers underline the disproportionate environmental burden of ASF and highlight the potential impact of changing diets on climate change.

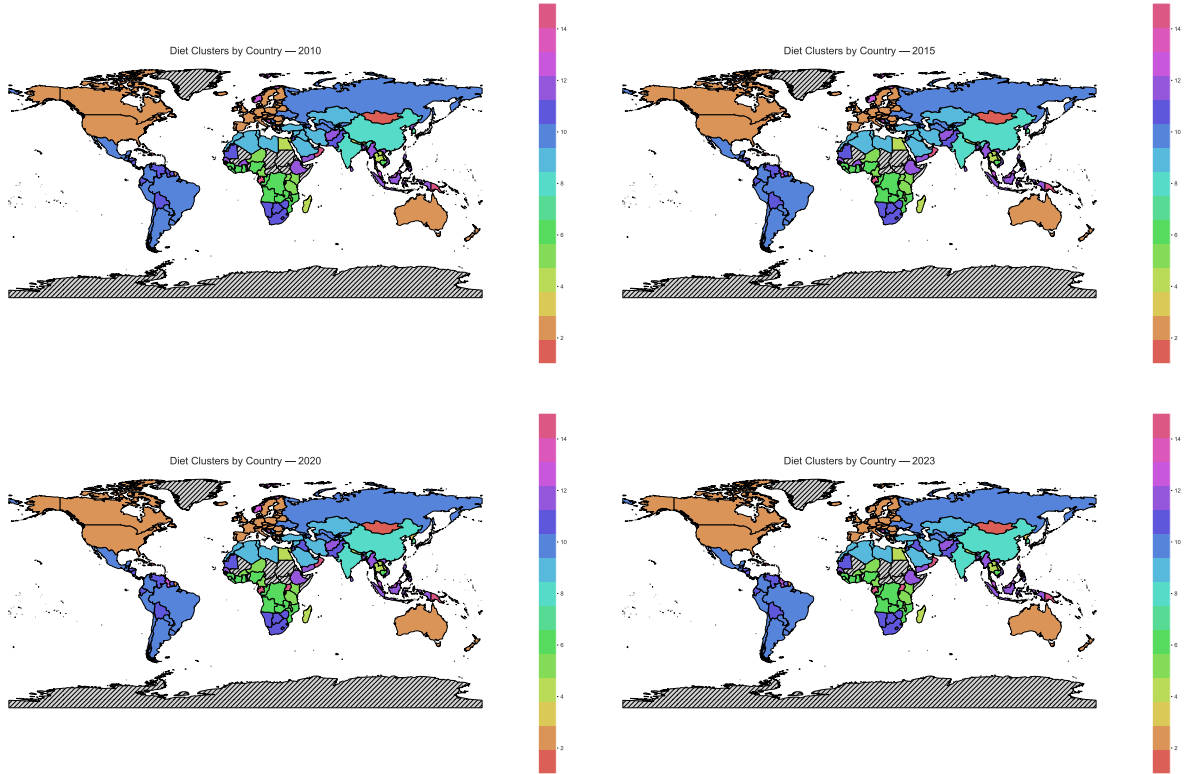


Figure 2: Change of dietary clusters over time.

Systematic similarities in dietary patterns across countries and over time, have been discovered with unsupervised hierarchical clustering. Based on the elbow criterion (Figure 4), 15 clusters are identified to be optimal. The resulting clusters in Figure 2 exhibit a high temporal stability, with relatively little change in countries' cluster assignments over the observation period. At the same time, the clusters display clear regional and cultural groupings. This persistence suggests that national dietary patterns are deeply embedded and slow to adjust.

Therefore, the goal of this study is to identify key factors that influence national dietary patterns, especially ASF consumption. To examine these factors, this study employs panel regression methods on a global country-year dataset covering the period 2010-2023.

2 Literature Review

Dietary behaviour has long been analysed through the lens of consumer theory, in which food choices are shaped by relative prices, income, and preferences subject to budget constraints [7]. Engel's law predicts that as income rises, the share of expenditure devoted to foods declines while demand for higher-value food products increases [8]. In nutritional science, dietary transitions are often described by the nutrition transition theory, which links rising incomes, urbanisation, and globalisation to increased consumption of animal-sourced and processed foods [21][27][19]. Cultural norms, religious factors, and historical food availability further shape persistent national dietary patterns [16].

2.1 Previous Empirical Results

Multiple reviews confirm that ASF exhibits particularly high emissions compared to PSF, contributing disproportionately to climate change and biodiversity loss [19]. Moreover, global modelling studies of healthy and sustainable diets conclude that shifts towards plant-based dietary patterns represent a cost-effective lever for achieving both environmental and public health objectives [22].

A large body of empirical literature links rising incomes, urbanisation, and structural socio-economic change with increased consumption of animal-sourced foods (ASFs). Using household-level data from rural China, Xu et al. document a *nutrition transition* associated with accelerating urbanisation, characterised by higher diet quality and increased consumption of animal-sourced and processed foods [27]. Global analyses of food demand further confirm that as countries become more affluent, share of staple foods like starchy roots declines and demand shifts towards more resource-intensive foods, including meat and dairy products [19]. Earlier work by York and Gossard uses data for 137 countries to examine cross-national meat and fish consumption, showing that indicators of modernisation (such as economic development and urbanisation) are positively associated with per capita meat intake, while ecological context also matters [28]. Together, these studies support the view that ASF consumption is shaped by macro-level socio-economic and demographic factors, which motivates the use of a

country-year panel framework in the present study.

At the cross-country level, Andreoli et al. analyse protein consumption patterns in 142 countries and find that per capita income is a key driver of animal-based protein consumption, alongside ecological, demographic, and social factors [1]. Other studies find meat consumption follows a Kuznets-distribution with an inflection point at an income of US\$36,375, which is quite high for any realistic impact [5].

2.2 Limitations of Previous Approaches

Despite the rich literature on the environmental impacts of ASFs, fewer studies link these impacts to the structural determinants of ASF consumption in a long-term cross-country framework. York and Gossard’s analysis is cross-sectional and focuses on a single point in time [28], while Andreoli et al. also estimate cross-sectional OLS regression for 2017 [1]. These studies identify important correlates of meat and protein consumption but do not capture temporal dynamics or persistence in national dietary patterns.

While a substantial number of household-level, national, and regional studies on ASF consumption exist, relatively few analyses combine long time horizons with broad cross-country coverage in a unified panel-data framework [27][1][28]. The existing literature provides strong evidence on the environmental burden of ASFs and on some cross-sectional determinants of meat and protein consumption, but offers limited insight into how structural macro-level factors shape the evolution of ASF-intensive dietary patterns across countries over time[20]. Policy-oriented reviews further note that many dietary recommendations rely heavily on assumptions derived from high-income countries, thereby limiting their global applicability [12].

2.3 Goal of This Research

This study contributes to the literature by combining unsupervised dietary clustering with panel regression analysis on a global dataset for 2010–2023. By linking stable dietary regimes to economic, demographic, and structural factors, the analysis provides insights into the macro-level drivers of ASF consumption and the scope for policy-induced dietary change. The strong temporal stability observed in the dietary clusters of Figure 2, identified in this study further

underscores the importance of analysing long-run structural drivers.

3 Theoretical Framework and Hypotheses

This study adopts a structural demand framework for animal-sourced food (ASF) consumption that combines consumer demand theory, the nutrition transition hypothesis, and the persistence of national dietary regimes. In line with standard consumer theory, food choices are determined by relative prices, income, and preferences subject to budget constraints [7]. Engel's law further predicts that rising incomes are associated with a declining budget share for staple foods and increasing demand for higher-value food products, including meat and dairy [8].

The nutrition transition theory posits that economic development, urbanisation, and global integration systematically shift dietary patterns towards higher consumption of animal-sourced and processed foods [21]. At the same time, cultural norms, historical food availability, and institutional structures create persistent national dietary patterns that adjust only gradually over time. The high temporal stability observed in the dietary clusters identified in this study provides empirical support for this persistence.

Compared with much of the existing literature, which relies predominantly on cross-sectional comparisons, the present framework explicitly accounts for both time-invariant country-specific factors and dynamic structural drivers by using a long-term country–year panel approach. By integrating persistent dietary regimes with macro-level economic and demographic determinants, the framework allows for a more robust identification of the long-run drivers of ASF consumption.

Based on this theoretical framework, the following testable hypotheses are derived.

Hypothesis 1 (Income Effect) *Higher GDP per capita is associated with a higher share of animal-sourced foods in total dietary energy consumption.*

Hypothesis 2 (Urbanisation Effect) *A higher degree of urbanisation increases ASF consumption due to shifts in lifestyles, food access, and dietary preferences.*

Hypothesis 3 (Price Effect) *Higher relative prices of animal-sourced foods are associated with lower ASF consumption shares.*

4 Data Sources and Variables

This study combines food supply data, socio-economic indicators, and price information from multiple international databases to construct a global country–year panel covering the period 2010–2023. The data sources are chosen to align with the theoretical framework outlined in section 3, which emphasises structural economic drivers, demographic structure, and price mechanisms. Income, price, and urbanisation capture demand-side mechanisms predicted by consumer theory and the nutrition transition. In addition, dietary clusters are identified through unsupervised hierarchical clustering of FAOSTAT food supply data [10]. These clusters could be used to do future cluster specific analysis, as clusters may have very different behaviours.

4.1 Dependent Variable

The primary dependent variable is the *share of animal-sourced foods (ASF)* in total dietary energy supply, measured in kilocalories. This variable captures the intensity of ASF consumption in national diets and directly reflects the outcome of interest in the theoretical framework, namely the structural demand for animal-based foods.

ASF calorie shares are computed using food balance sheet data from the Food and Agriculture Organization (FAO) FAOSTAT database [10]. Food items are classified into animal- and plant-sourced categories, and total caloric availability is aggregated at the country–year level. The resulting ASF share represents a consistent and internationally comparable proxy for national dietary patterns.

4.2 Independent Variables

The main independent variables are derived from the World Bank World Development Indicators (WDI) and are intended to capture the structural drivers of dietary change highlighted in consumer demand theory and nutrition transition theory [26].

GDP per capita (constant US dollars) is used as a proxy for income levels, reflecting Engel’s law and the expected positive relationship between income growth and demand for higher-value food products[8]. Urbanisation is measured as the share of the population living in urban

areas and captures lifestyle changes, market access, and dietary diversification associated with economic development.

To account for price effects in food demand, the analysis incorporates food price indicators from two sources. First, the FAO Food Price Index provides a global measure of international food price developments over time [9]. Second, country-level consumer food price data is obtained from the FAOSTAT Prices and Producer Prices database [11]. These variables capture absolute price movements in local currency that influence food choices within the consumer demand framework for individual items. From a basket of weighted products a price index scales 0-1 for every country is computed. 1 being the highest price of the ASF and PSF baskets, and 0 representing 0 in the local currency.

4.3 Control Variables

Additional macroeconomic controls, including trade openness, demographic structure, access to electricity and population size, are obtained from the World Bank API and included to isolate the effect of income and urbanisation from broader economic trends. Demographic structure is proxied by the share of the population aged 25 and younger, reflecting cohort effects consumption habits.

4.4 Time-Invariant Variables

Several variables relevant to dietary patterns are largely time-invariant over the study period and therefore cannot be included directly in the panel regression framework with country fixed effects. These include dominant religious affiliation, colonial history, and climatic conditions. Data on dominant religion are sourced from Our World in Data [18], colonial history indicators from the CEPII colonial relationship database [3], and climate classifications from Weather and Climate [23].

While these variables are excluded from the fixed-effects regressions, they could be used descriptively to interpret dietary clusters and regional patterns. In particular, they could contextualise persistent dietary regimes identified in the clustering analysis and highlight structural constraints that may limit short-run dietary change.

5 Methods

This study employs panel regression methods to examine the structural determinants of animal-sourced food (ASF) consumption using a global country–year dataset covering the period 2010–2023. The empirical approach is guided by section 3, which emphasises income effects, price mechanisms, and demographic structure.

5.1 Panel Regression Framework

Panel data methods are particularly well suited to the analysis of dietary patterns for several reasons. First, panel regression allows the analysis to control for unobserved, time-invariant country-specific factors—such as cultural norms, historical food systems, and long-run climatic conditions, which are likely to influence ASF consumption but are difficult to measure directly[2]. By incorporating country fixed effects, the model isolates the impact of time-varying economic and demographic variables on dietary outcomes.

Second, panel data exploit both cross-sectional and temporal variation, improving estimation efficiency and reducing omitted-variable bias relative to purely cross-sectional approaches [14]. This is particularly important given the strong persistence observed in national dietary patterns over time.

Third, panel regression enables the modelling of dynamic adjustment processes, allowing current dietary outcomes to depend on past consumption levels. Such persistence is consistent with path dependence in food systems and long-run dietary habits [25]. Accounting for these dynamics is essential when analysing gradual changes in national diets.

5.2 Model Specification

The baseline empirical specification is given by:

$$ASF_{it} = \alpha_i + \beta X_{it} + \gamma P_{it} + \delta C_{it} + \varepsilon_{it},$$

where ASF_{it} denotes the share of animal-sourced foods in total dietary energy consumption in country i at time t , α_i captures country fixed effects, X_{it} represents the main independent variables derived from the theoretical framework (income, urbanisation, and demographic structure), P_{it} includes food price indicators, and C_{it} denotes additional control variables.

All models are estimated using fixed-effects estimators to account for unobserved heterogeneity across countries. Time fixed effects are included to control for global shocks common to all countries, such as international food price fluctuations or macroeconomic disturbances. Standard errors are clustered at the country level to allow for serial correlation within panels.

5.3 Descriptive Analysis

Unsupervised hierarchical clustering of dietary patterns is used exclusively for descriptive and visualisation purposes to illustrate similarities and persistence in national diets across countries and over time. Importantly, clustering results are not included as explanatory variables in the regression analysis to avoid mechanical correlation and target leakage, as dietary clusters are constructed directly from the dependent variable.

6 Results

Table 1: Determinants of Animal-Sourced Food Consumption

Dep. Variable:		animal_share	R-squared:	0.1472		
F-statistic:		25.952	P-value	0.0000		
	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
	log_gdp_pc	0.0502	3.9448	0.0001	0.0252	0.0751
	asf_price_index	0.0010	0.5083	0.6113	-0.0028	0.0048
	psf_price_index	0.0069	1.3951	0.1632	-0.0028	0.0166
	young_share	0.0249	1.9769	0.0483	0.0002	0.0497
	urban_pop_share	-0.0223	-1.2926	0.1964	-0.0563	0.0116
	arable_land_km2	0.0207	1.8122	0.0702	-0.0017	0.0431
	Population	0.0167	2.1510	0.0317	0.0015	0.0318
	access_electricity	-0.0059	-1.1931	0.2331	-0.0156	0.0038
	trade_openness	-0.0025	-0.8329	0.4050	-0.0085	0.0034
	const	0.1981	9.928e+15	0.0000	0.1981	0.1981

Table 1 and Table 2 report the results of the fixed-effects panel regressions examining the determinants of animal-sourced food (ASF) consumption. Across all specifications, GDP per capita emerges as the most robust and economically meaningful predictor of ASF consumption.

In the baseline specification Table 1, the coefficient on log GDP per capita is positive and statistically significant ($p=0.0001$). The estimated magnitude implies that a 10% increase in GDP per capita is associated with an increase of approximately 0.5 percentage points in the share of animal-sourced foods in total dietary energy. This finding is consistent with Engel's law and the nutrition transition hypothesis, which predict higher consumption of animal-based foods as incomes rise.

Demographic structure also plays a role in shaping dietary patterns. The share of young people in the population is positively associated with ASF consumption and somewhat statistically significant ($p=0.0710$). This suggests that younger populations tend to consume more ASFs, potentially reflecting dietary preferences, or differences in nutritional requirements.

By contrast, most other covariates are not statistically significant at conventional levels. Urbanisation, trade openness, access to electricity, and arable land availability do not exhibit robust associations with ASF consumption once country fixed effects are included. Similarly, food price indicators show no statistically significant effects, indicating that short-run price variation has limited explanatory power for long-run national dietary patterns at the global level.

Table 2: Determinants of Animal-Sourced Food Consumption on log price index ratios

Dep. Variable: animal_share		R-squared: 0.1349			
F-statistic: 30.193		P-value 0.0000			
	Parameter	Std. Err.	P-value	Lower CI	Upper CI
log_gdp_pc	0.0406	0.0141	0.0040	0.0129	0.0683
log_rel_price	0.0157	0.0493	0.7509	-0.0811	0.1124
log_rel_price_x_log_gdp	-0.1496	0.4401	0.7340	-1.0130	0.7138
young_share	0.0025	0.0012	0.0401	0.0001	0.0048
urban_pop_share	-0.0011	0.0008	0.1857	-0.0028	0.0005
arable_land_km2	6.6436	4.1560	0.1102	-1.5092	14.796
log_pop	-0.0161	0.0191	0.3999	-0.0536	0.0214
const	-0.0587	0.2714	0.8287	-0.5911	0.4737

To further analyse the relationship of price and ASF consumption another model is employed: Table 2, which includes relative food prices and an interaction term between prices and income. It strengthens previous findings. While GDP per capita and demographic structure are statistically significant, neither relative prices nor their interaction with income exert a meaningful influence on ASF consumption. This suggests that income effects dominate price effects in explaining cross-country variation in ASF consumption over the study period.

Overall, the results indicate that ASF consumption is primarily driven by long-run structural factors. Most notably income levels and demographic composition, rather than by short-term economic fluctuations or price movements. Therefore accepting hypothesis 1 while not being able to proof hypothesis 2 and 3. Future analysis could and should focus on analysing the impact of youth population on ASF consumption.

7 Appendix

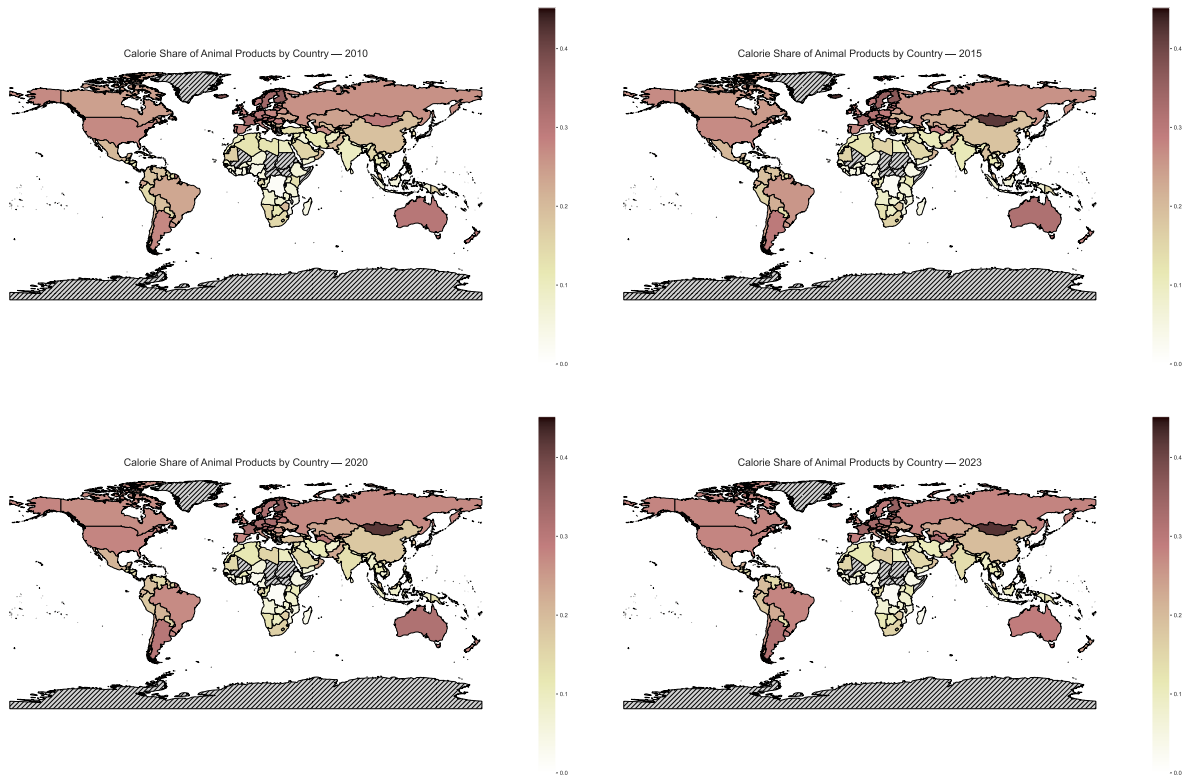


Figure 3: Change of animal-sourced foods share over time.

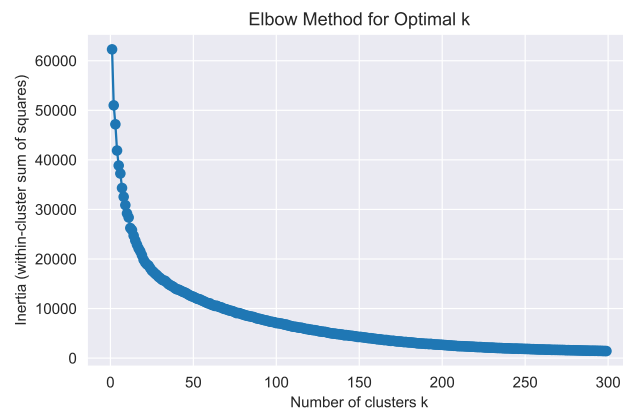


Figure 4: Elbow plot for 1 to 300 clusters.

Table 3: Determinants of Animal-Sourced Food Consumption

Dep. Variable:	animal_share	R-squared:	0.1472
Estimator:	PanelOLS	R-squared (Between):	-0.0068
No. Observations:	1486	R-squared (Within):	0.1343
Date:	Fri, Dec 26 2025	R-squared (Overall):	-0.0070
Time:	16:00:12	Log-likelihood	4353.7
Cov. Estimator:	Clustered		
		F-statistic:	25.952
Entities:	111	P-value	0.0000
Avg Obs:	13.387	Distribution:	F(9,1353)
Min Obs:	1.0000		
Max Obs:	14.000	F-statistic (robust):	6.0111
		P-value	0.0000
Time periods:	14	Distribution:	F(9,1353)
Avg Obs:	106.14		
Min Obs:	104.00		
Max Obs:	109.00		

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
log_gdp_pc	0.0502	0.0127	3.9448	0.0001	0.0252	0.0751
asf_price_index	0.0010	0.0020	0.5083	0.6113	-0.0028	0.0048
psf_price_index	0.0069	0.0049	1.3951	0.1632	-0.0028	0.0166
young_share	0.0249	0.0126	1.9769	0.0483	0.0002	0.0497
urban_pop_share	-0.0223	0.0173	-1.2926	0.1964	-0.0563	0.0116
arable_land_km2	0.0207	0.0114	1.8122	0.0702	-0.0017	0.0431
Population	0.0167	0.0077	2.1510	0.0317	0.0015	0.0318
access_electricity	-0.0059	0.0049	-1.1931	0.2331	-0.0156	0.0038
trade_openness	-0.0025	0.0030	-0.8329	0.4050	-0.0085	0.0034
const	0.1981	1.995e-17	9.928e+15	0.0000	0.1981	0.1981

F-test for Poolability: 183.50

P-value: 0.0000

Distribution: F(123,1353)

Included effects: Entity, Time

Table 4: Determinants of Animal-Sourced Food Consumption

Dep. Variable:	animal_share	R-squared:	0.1349
Estimator:	PanelOLS	R-squared (Between):	0.1666
No. Observations:	1486	R-squared (Within):	0.0552
Date:	Thu, Dec 18 2025	R-squared (Overall):	0.1788
Time:	11:10:39	Log-likelihood	4343.1
Cov. Estimator:	Clustered		
		F-statistic:	30.193
Entities:	111	P-value	0.0000
Avg Obs:	13.387	Distribution:	F(7,1355)
Min Obs:	1.0000		
Max Obs:	14.000	F-statistic (robust):	3.5944
		P-value	0.0008
Time periods:	14	Distribution:	F(7,1355)
Avg Obs:	106.14		
Min Obs:	104.00		
Max Obs:	109.00		

	Parameter	Std. Err.	T-stat	P-value	Lower CI	Upper CI
log_gdp_pc	0.0406	0.0141	2.8799	0.0040	0.0129	0.0683
log_rel_price	0.0157	0.0493	0.3175	0.7509	-0.0811	0.1124
log_rel_price_x_log_gdp	-0.1496	0.4401	-0.3399	0.7340	-1.0130	0.7138
young_share	0.0025	0.0012	2.0551	0.0401	0.0001	0.0048
urban_pop_share	-0.0011	0.0008	-1.3241	0.1857	-0.0028	0.0005
arable_land_km2	6.6436	4.1560	1.5986	0.1102	-1.5092	14.796
log_pop	-0.0161	0.0191	-0.8421	0.3999	-0.0536	0.0214
const	-0.0587	0.2714	-0.2164	0.8287	-0.5911	0.4737

F-test for Poolability: 172.83

P-value: 0.0000

Distribution: F(123,1355)

Included effects: Entity, Time

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